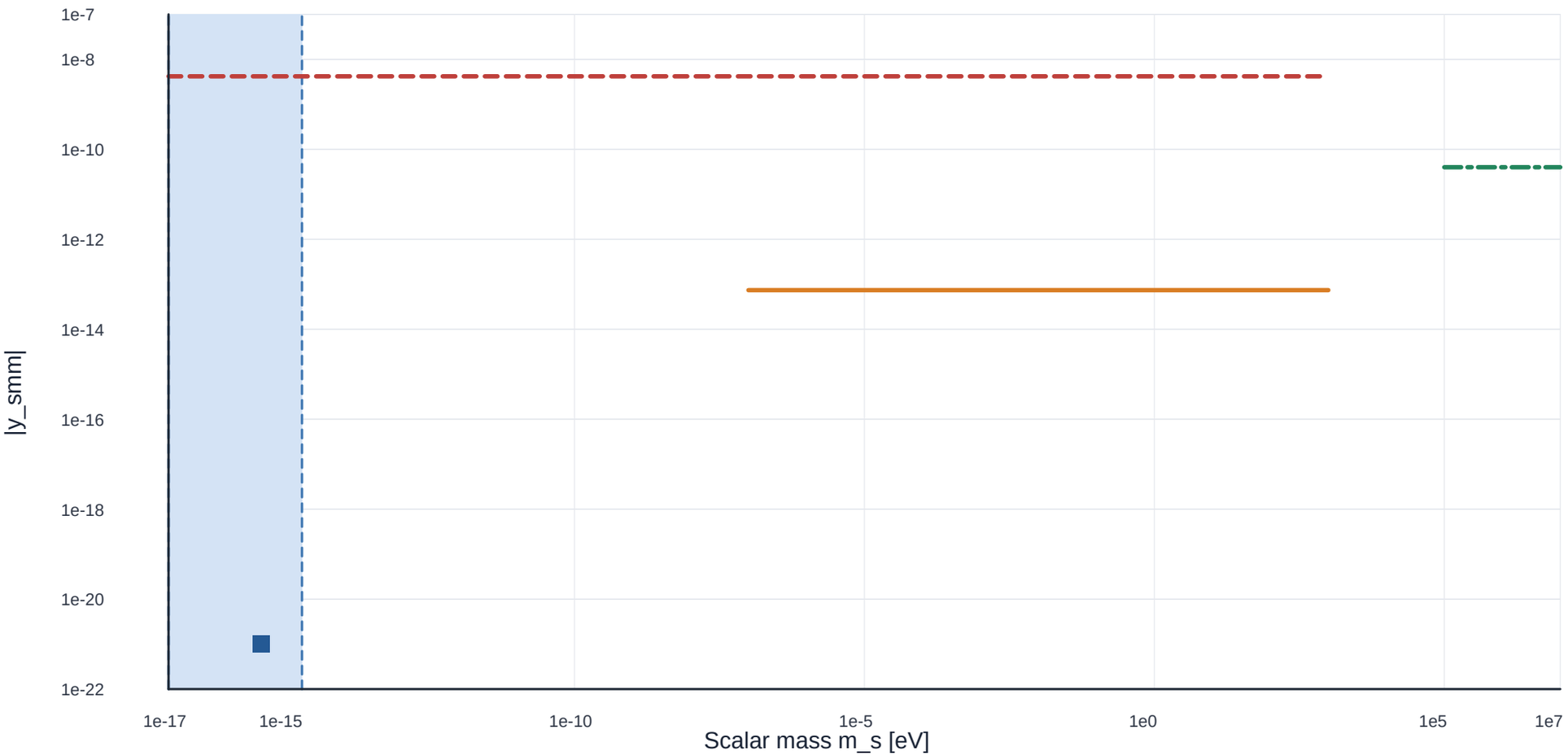
Scalar-muon Yukawa constraints

Source-anchored schematic summary - NOT digitized combined exclusion contours



- Binary-pulsar periastron advance: marker = best point (~1e-21 at 4e-16 eV); shaded interval = strongest sensitivity.
- Neutron-star cooling: |y_smm| ~7.4e-14 for m_s <= 1 keV (Fiorillo et al., Table 1).
- SN 1987A cooling: ~4.2e-9 for m_s <= 1 keV; free-streaming threshold only, not full trapping/explosion exclusion.
- Diffuse SN gamma rays: ~4e-11 for m_s >= 0.1 MeV; plateau to 10 MeV (Caputo et al., Fig. 11).

Caution: Segments show source-reported regimes only; no interpolation or full excluded band is inferred.

Scalar limits act on y_smm; mapping to (m_A', g_X) is conditional on the stated UV benchmark.